

A ROS-Based Multimodal Interaction Framework for Expressive and Adaptive Social Robots

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Abstract—This paper introduces a unified multimodal communication architecture for a compact social robot used in autism therapy. The system combines speech, touch, gestures, pointing, and gaze cues within a single perception framework, and coordinates expressive behaviours through a centralized Interaction Manager. Multimodal inputs are normalized into unified intents, while expressive output such as emotional TTS, LEDs, face expressions, head motion, gaze, and vocalizations, are synchronized through a single interface. The architecture enables coherent, predictable behaviours that support therapeutic activities requiring coordinated multimodal interaction.

I. INTRODUCTION

Social robots designed for child–robot interaction increasingly rely on multimodal communication: speech, touch, gestures, facial expressions, head motion, and gaze. However, these modalities are often implemented as independent sub-systems, resulting in asynchronous or conflicting behaviors that reduce interaction quality. This challenge is particularly relevant in therapeutic contexts, where children with Autism Spectrum Disorder (ASD) benefit from predictable timing, clear expressive cues, and consistent multimodal feedback. Social robots have been widely explored as tools to support ASD interventions [4].

This paper presents a unified multimodal communication architecture for a compact, low-cost social robot used in ASD therapy. The robot builds on our previous work on socially assistive platforms for autism therapy [1], [2].

The contribution of this paper is not the robot hardware itself, but a **Interaction Manager** that coordinates perception, intent extraction, and expressive robot behaviors into a coherent interaction pipeline. The system integrates speech recognition, touch input, gesture and pointing detection, expressive LED-based facial animation, emotional text-to-speech, head motion, and gaze control. Gaze is treated as an expressive output modality and as a perceptual state signal for estimating engagement, but not as an intentional user input. By synchronizing these modalities through a single abstraction layer, the robot produces natural, temporally aligned behaviors that support therapeutic engagement.

We describe the robot platform, the multimodal perception pipeline, the expressive output system, and the Interaction Manager that unifies them. We then illustrate how this ar-

chitecture enables therapeutic activities requiring coordinated multimodal interaction.

II. ROBOT PLATFORM OVERVIEW

A. Hardware

The Emy robot (Figure 1) is a compact tabletop platform (20–30 cm) equipped with a three-degree-of-freedom head (yaw, roll, pitch), a full LED-matrix expressive face, RGB body LEDs, a frontal camera, an IMU, and multiple touch sensors located on the face, shoulders, and base. The onboard computation is provided by an NVIDIA Jetson Orin, while computationally intensive modules such as ASR, TTS, and LLM-based dialogue run on an external workstation. It also comes with an external touch-screen used for multimedia interaction.

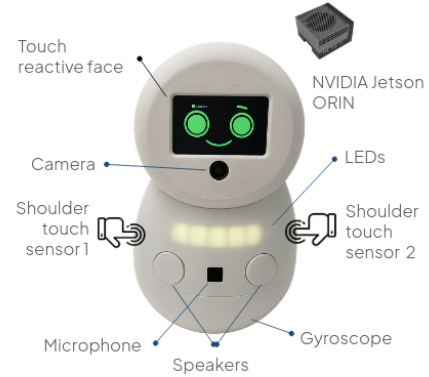


Fig. 1. EMy robot

III. MULTIMODAL INPUTS

Our multimodal perception stack builds on ROS4HRI [5], a standardized framework for representing humans, faces, voices, and social cues in ROS 2. ROS4HRI provides unified entities and topic structures for multi-user interaction, enabling consistent integration of speech activity, face tracking, head gestures, and gaze-related cues across the different modalities described in the following

A. Speech Input

Speech is processed using Vosk ?? ASR models optimized for Spanish. The system is robust to common characteristics of children’s speech, including low volume and variable pronunciation. They are captured through ROS4HRI’s /humans/voices/anonymous_speaker/speech.

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B. Touch Input

Touch events originate from two sources: the robot's physical touch sensors (face, shoulders, base) and the external touchscreen interface. Physical touches are published on a dedicated `/robot/touch` topic and are treated as social cues that may trigger affective reactions (e.g., a shoulder touch prompting a brief positive acknowledgement). Touchscreen interactions are published on `/ui/input` and represent functional UI commands.

C. Vision-based Perception

The robot uses an RGB camera as the main visual sensor for interaction perception and scene understanding. The overall perception framework follows our previously developed real-time social vision architecture [3] which integrates with ROS4HRI entities for representing faces, identities, and head gestures.

Face detection is done using YuNet [7], a lightweight detector that provides efficient real-time localisation of participants in the scene. Detected faces are continuously tracked over time, allowing the system to maintain stable identities. Moreover, detected face images are processed using dlib-based facial embeddings for face identification.

In parallel, pointing recognition is estimated from upper-body and hand landmarks provided by the MediaPipe models [9]. We infer the user's pointing direction in real time with a geometric approach from these landmarks. By combining both sources of information, hand and body, the robot can estimate which object or region of the scene is being referenced.

Facial landmarks are also used to estimate engagement through gaze direction and head orientation cues. These signals indicate whether a user is attending to the robot, another person, or an object of interest, and can be used to adapt timing, turn-taking, or attention-recovery behaviours.

The same landmark information is used for head gesture recognition, detecting affirmative and negative responses through nodding and shaking motions. This geometric approach avoids the need for heavier learning-based models while remaining suitable for real-time execution

IV. MULTIMODAL OUTPUTS

A. Expressive Face

The expressive face supports more than a dozen predefined expressions, including Happy, Sad, Skeptical, Thinking, and Sleepy. Each expression specifies eye and brow shapes, mouth curvature, color, and optional animations. The system also supports highlighting specific regions (eyes, brows, mouth), changing the color of the face, and smooth gaze control via a ROS2 topic `/robot_face/expression`.

B. Head Motion and Gaze Control

Head motion and gaze are organised as a layered ROS 2 control architecture composed of two behaviour sources: continuous policies and predefined motions. Policies provide ongoing gaze behaviours that update eye and head targets



Fig. 2. EMy robot's facial expressions

during interaction, whereas predefined motions are one-time scripted actions such as nodding or shaking the head.

Policy-based behaviours are coordinated by the `gaze_manager_node`, which receives high-level gaze requests from the decision-making layer, selects the active policy, and determines the 3D target to observe. For face-based behaviours, it uses detections from the perception stack and selects the corresponding face position as the gaze target. When no face is available, it can perform search or exploration behaviours. Examples of policies include:

- `social`: track detected human faces,
- `reset`: return the head and eyes to a neutral pose,
- `explore_face`: search for faces and lock onto one when detected.

Most policies operate with priority level 50, while the `reset` policy is assigned priority 100 to guarantee immediate return to neutral when required. The selected target is forwarded to the `robot_look_at` node, which converts the 3D point into coordinated eye and neck movements. Small deviations are handled by the eyes alone, while larger offsets engage combined head motion to support joint attention, turn-taking, and subtle expressive cues.

Predefined communicative motions are handled separately through the `motion_executor_node`, which triggers scripted sequences such as nodding (yes) or shaking the head (no). These motions are assigned priority level 80, allowing them to temporarily override active gaze policies when necessary.

At the lowest level, all neck trajectories are executed by the ROS 2 `joint_trajectory_controller`. Based on command priority, it selects the active trajectory source and generates smooth, time-parameterised motion for the neck servos while enforcing joint limits.

C. Speech Synthesis

The robot uses an external Text-to-Speech server running on the workstation to generate expressive speech. The system is based on a Dockerized Coqui XTTS model, which provides multilingual neural synthesis with controllable emotional styles (e.g., *happy*, *surprised*, *sad*, *afraid*) and supports whisper-like voices for calming or low-intensity interaction. Emotion control is achieved through a voice-cloning model trained on curated emotional speech samples.

A ROS 2 action interface (/say) exposes the TTS capabilities to the Interaction Manager, allowing modules to specify language, emotion, volume, and sampling temperature.

D. Vocalization

The robot can play any WAV audio file and has an extensive set available such as "sigh", "yawn", "cry", "shout", "celebrate".

V. THE INTERACTION MANAGER

A. Motivation

Multimodal inputs arrive asynchronously, and expressive outputs must be coordinated to avoid unnatural or conflicting behaviors. The Interaction Manager addresses this by providing a unified abstraction layer that manages intent extraction, dialogue flow, and synchronized multimodal output.

The Interaction Manager provides a unified abstraction layer that connects multimodal perception with expressive robot behaviour. It is organised around three core functions:

B. Intent Normalization

Intentional user actions (speech, touch, gestures, pointing) are normalized into a unified /intent message containing a JSON payload and a modality flag, which can be either ASR, touchscreen, gesture. This simplifies integration with dialogue managers, LLMs, and behavior planners (Figure 3). Continuous signals such as gaze and engagement are handled separately to avoid polluting the intent stream.

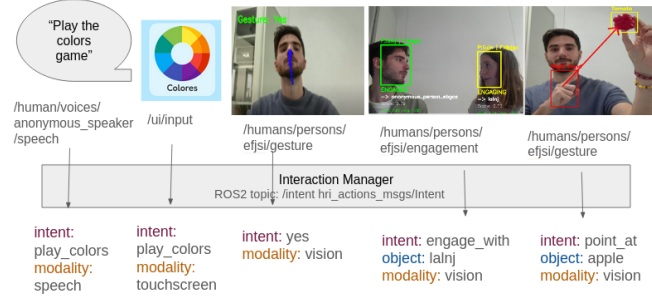


Fig. 3. Multimodal input aggregator to produce intents

C. Multimodal Output Coordination

Multimodal output is coordinated by the Interaction Manager through the /expressive_say ROS 2 action interface. This interface synchronizes combinations of TTS (including its emotional prosody and content), facial expressions, LEDs, head motion, and non-verbal vocalizations from a single annotated utterance, by using inline tags for each modality.

This provides a compact and expressive mechanism for generating coherent, temporally aligned robot behaviours, preventing overlapping actions.

It also supports triggering of predefined multimodal reactions stored in a YAML configuration file (robot_behaviors.yaml). Each reaction specifies the coordinated execution of TTS emotion, facial expression,

LED effects, head motion, and optional vocalizations. These definitions allow the system to trigger rich, synchronized behaviors through a single tag such as <reaction(Happy)>.

An example configuration entry is shown below.

```
Happy:
  description: "The robot
  expresses happiness."
  tts:
    prosody: "happiness"
  face: "Happy"
  leds:
    effect: "color_cycle"
    colors: ["green", "yellow", "magenta"]
    speed: 0.5
    cycles: 3
  motion:
    name: "nod_fast"
    timing: "before_tts"
  vocalization:
    category: "soft_laugh"
```

A typical invocation from the command line is shown below:

```
ros2 action send_goal /expressive_say \
  audio_tts_msgs/action/Communication \
  "text: '<reaction(Happy)>Good to see you!'
  <do motion(nod)>'
  language: es"
```

Supported Inline Tags: The Expressive Say Engine supports the following inline tags, which are parsed sequentially and executed by the Interaction Manager:

- **<expression(NAME)>** Sets the robot's facial expression for the following text. *Example:* <expression(Happy)>Hola!
- **<tts(EMOTION)>** Selects the TTS emotional style. Supported emotions include neutral, happy, sad, upset, whisper. *Example:* <tts(whisper)>habla bajito...
- **<do motion(NAME)>** Executes a predefined head motion *before* continuing speech (blocking). *Example:* <do motion(no)>No, no, no...
- **<set motion(NAME)>** Starts a non-blocking head motion that runs in parallel with speech. *Example:* <set motion(nod)>sí...
- **<reaction(NAME)>** Triggers a multimodal reaction defined in the YAML configuration (TTS emotion, face, LEDs, motion, vocalization). *Example:* <reaction(Happy)>Qué bien!

D. Dialogue Management

When enabled, the Hub forwards speech inputs to an LLM-based chatbot running on an external workstation. Strict turn-taking rules prevent interruptions, and contextual information is added to maintain coherence across interaction turns.

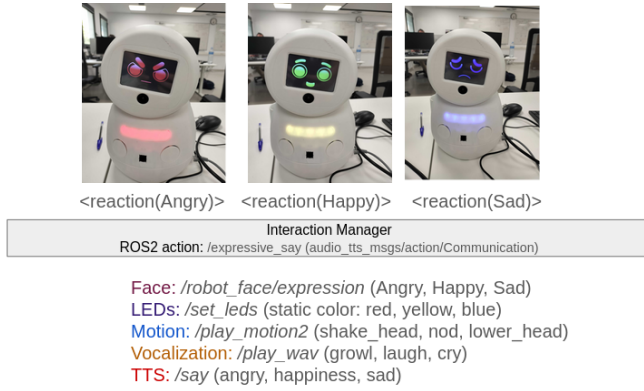


Fig. 4. Reactions generation

VI. USE CASES IN THERAPEUTIC INTERACTION

The robot is used in therapeutic sessions with children diagnosed with Autism Spectrum Disorder (ASD), where predictable timing, clear expressive cues, and multimodal feedback help sustain engagement and support communication goals. Activities typically require the robot to combine several modalities such as speech, touch, pointing, gaze, and expressive behaviours, in order to create structured yet flexible interaction patterns that children can easily follow.

The multimodal architecture explained in this paper is used for example in these two contexts:

Multimodal Vocabulary Learning: The robot integrates several vocabulary learning games, where it presents a concept such as a colour or facial expression words. For example, to teach expression vocabulary, the robot uses a multimodal reaction to trigger an expression such as sadness. It then asks the child a question such as “Am I feeling sad?”. The child may respond verbally (ASR), by selecting or pointing an option on the touchscreen, or through head gestures indicating yes/no. All these inputs are captured as normalized /intent messages (differing on modality type), allowing the activity to proceed consistently regardless of the modality chosen by the child (see Figure 5). This flexibility is particularly important in ASD therapy, where children may prefer different communication channels.

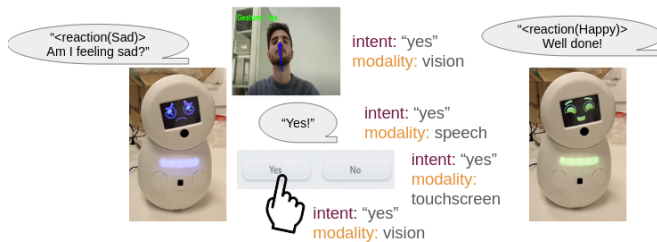


Fig. 5. Multimodal architecture applied on an expression vocabulary game

Adaptive Reactiveness Based on Engagement: During the therapy session, the robot monitors the child’s engagement through gaze direction and head orientation cues. When disengagement is detected, the robot adjusts its gaze policy.

For example, alternating its attention between the child and the touchscreen to re-establish joint attention. Physical touch also plays a role: touching the robot’s shoulder triggers a positive acknowledgement, while touching the face produces a blink reaction. Future work will integrate these tactile cues more deeply into the decision-making layer, enabling richer adaptive behaviours.

VII. CONCLUSION

We presented a unified multimodal communication architecture that integrates perception, intent extraction, and expressive robot behaviours for a compact, low-cost social robot used in ASD therapy. By coordinating speech, gaze, facial expressions, LEDs, head motion, and non-verbal vocalizations through a single Interaction Manager, the system produces natural, synchronized behaviours suitable for therapeutic contexts.

A separate decision-making module is work in progress that uses the unified intent stream and perceptual cues to select appropriate robot actions, enabling more adaptive and context-sensitive interaction.

Future work will focus on expanding the set of multimodal reactions based on feedback from children and therapists, and on further integrating engagement-driven behaviours to support richer, more responsive therapeutic interactions.

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